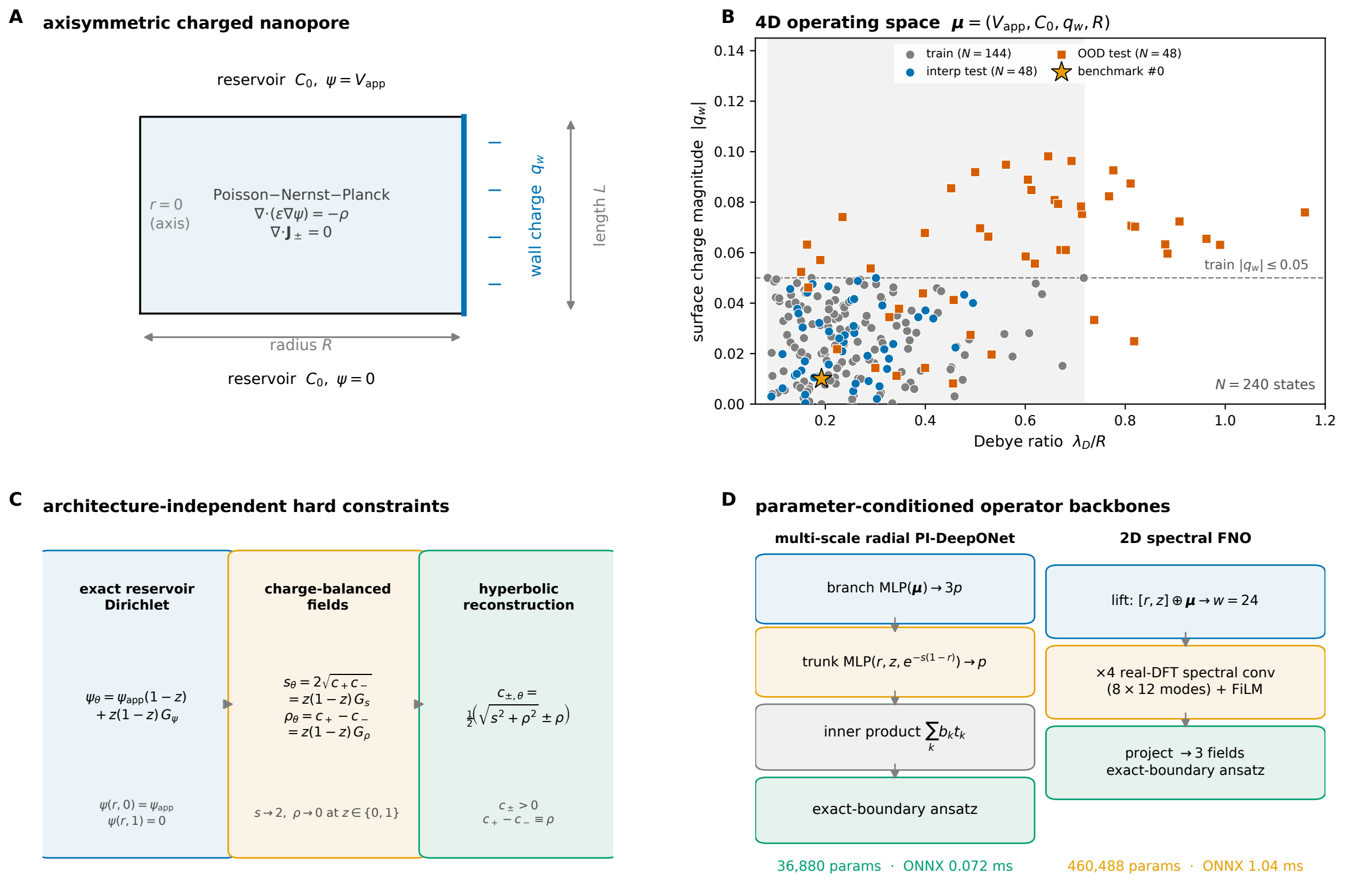

A
axisymmetric charged nanopore
reservoir C_0, $\psi = V_{\text{app}}$
$r = 0$
(axis)
Poisson−Nernst−Planck
$\nabla\cdot(\varepsilon\nabla\psi) = -\rho$
$\nabla\cdot\mathbf{J}_\pm = 0$
wall charge q_w
length L
radius R
reservoir C_0, $\psi = 0$
B
4D operating space $\boldsymbol{\mu} = (V_{\text{app}}, C_0, q_w, R)$
train ($N = 144$)
interp test ($N = 48$)
OOD test ($N = 48$)
benchmark #0
surface charge magnitude $|q_w|$
train $|q_w| \le 0.05$
$N = 240$ states
Debye ratio λ_D/R
C
architecture-independent hard constraints
exact reservoir Dirichlet
$\psi_\theta = \psi_{\text{app}}(1-z) + z(1-z)\,G_\psi$
$\psi(r,0) = \psi_{\text{app}}$
$\psi(r,1) = 0$
charge-balanced fields
$s_\theta = 2\sqrt{c_+ c_-} = z(1-z)\,G_s$
$\rho_\theta = c_+ - c_- = z(1-z)\,G_\rho$
$s \to 2$, $\rho \to 0$ at $z \in \{0,1\}$
hyperbolic reconstruction
$c_{\pm,\theta} = \frac{1}{2}\left(\sqrt{s^2+\rho^2} \pm \rho\right)$
$c_\pm > 0$
$c_+ - c_- \equiv \rho$
D
parameter-conditioned operator backbones
multi-scale radial PI-DeepONet
branch MLP($\boldsymbol{\mu}$) → $3p$
trunk MLP($r, z, e^{-s(1-r)}$) → p
inner product $\sum_k b_k t_k$
exact-boundary ansatz
36,880 params · ONNX 0.072 ms
2D spectral FNO
lift: $[r,z] \oplus \boldsymbol{\mu} \to w = 24$
×4 real-DFT spectral conv (8 × 12 modes) + FiLM
project → 3 fields exact-boundary ansatz
460,488 params · ONNX 1.04 ms